

# CIS 421 Database Management Systems Final Project

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## Conceptual Model

- **Objective:** Design an **E-R (Entity-Relationship) schema** for your application.
    - **Tool:** You can use the Designer tool in phpMyAdmin (available under the "Designer" option in the Right Panel) or design the diagram using MS Word or MS PowerPoint.
  - **Content:** Ensure the E-R schema diagram accurately represents entities, attributes, and relationships.
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## Relations & Attributes

### PRODUCTS

*Primary Key:* (**Product\_ID**)

*Foreign Key(s):* N/A

- **Product\_ID**
- Name
- Price
- Stock

### CUSTOMERS

*Primary Key:* (**Customer\_ID**)

*Foreign Key(s):* N/A

- **Customer\_ID**
- Fname
- LName
- Phone
- Email

### ORDERS

*Primary Key:* (**Order\_Id**)

*Foreign Key(s):* *Cstmr\_ID* (FK: Customers.Customer\_ID)

- **Order\_Id**
- *Cstmr\_ID*
- Order\_date
- Due\_date
- Total\_amount

## ORDER\_ITEMS

Primary Key: (Ord\_ID, Prod\_ID)

Foreign Key(s): Ord\_ID (FK: Orders.Order\_ID), Prod\_ID (FK: Products.Product\_ID)

- Ord\_ID
- Prod\_ID
- Quantity

## INGREDIENTS

Primary Key: (Ingredient\_ID)

Foreign Key(s): N/A

- Ingredient\_ID
- Name
- Quantity
- Unit

## PRODUCT\_INGREDIENTS

Primary Key: (Prod\_ID, Ingr\_ID)

Foreign Key(s): Prod\_ID (FK: Products.Product\_ID), Ingr\_ID (FK: Ingredients.Ingredient\_ID)

- Prod\_ID
- Ingr\_ID
- Req\_Qty

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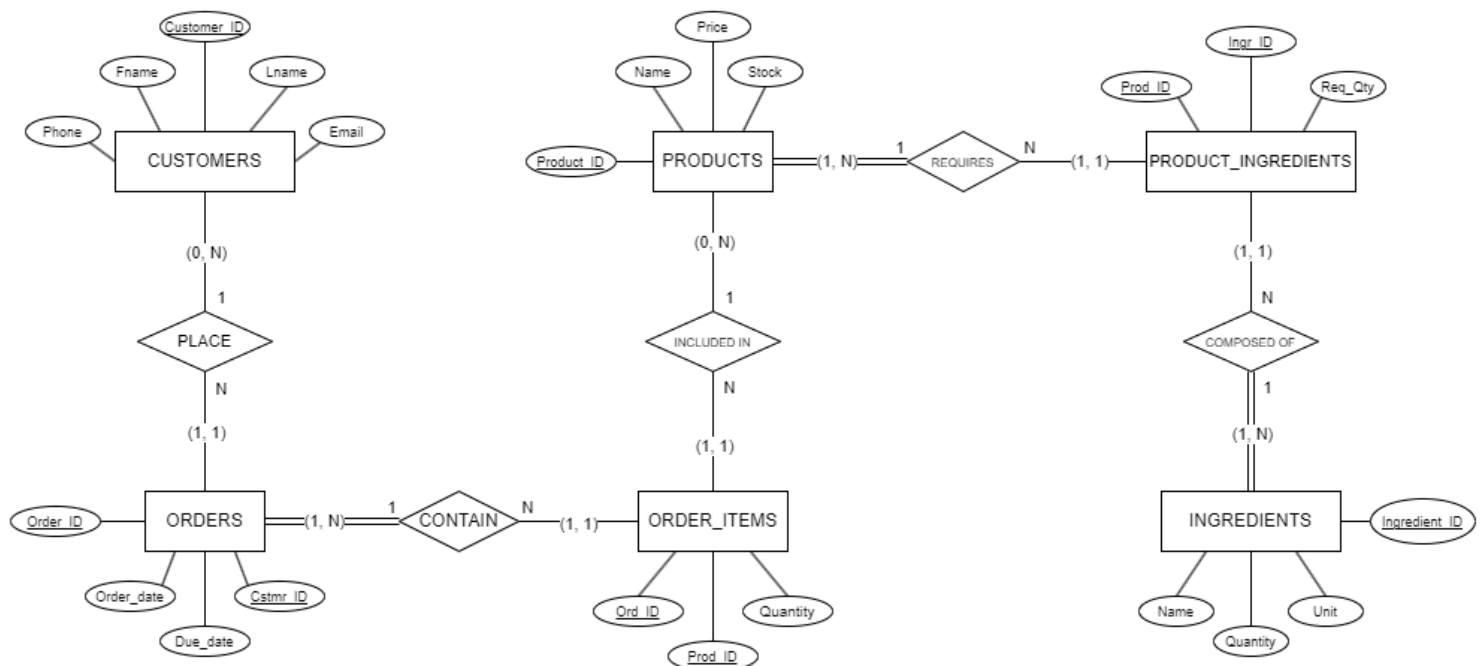
## Constraints on Participation

1. **PRODUCTS & ORDER\_ITEMS (1:N)** — Each product can appear in multiple order items, but each order item references a single product.
  - **PRODUCTS (0, N)** — a product can be an *order\_item* for many different orders or none at all.
  - **ORDER\_ITEMS (1, 1)** — every order\_item always references a single product sold by the bakery.
2. **CUSTOMERS & ORDERS (1:N)** — A customer can place multiple orders, but each order is made by a single customer.
  - **CUSTOMERS (0, N)** — a customer can place any number of orders or none.
  - **ORDERS (1, 1)** — every order is placed by exactly one customer.
3. **ORDERS & ORDER\_ITEMS (1:N)** — Each order can have multiple items, but every order item belongs to a single order.
  - **ORDERS (1, N)** — every order must contain at least one item but can also have many.
  - **ORDER\_ITEMS (1, 1)** — each order item is linked to exactly one order.
4. **PRODUCTS & PRODUCT\_INGREDIENTS (1:N)** — Each product can require multiple ingredients, but each entry in the **PRODUCT\_INGREDIENTS** table refers to a specific ingredient for a specific product.

- **PRODUCTS (1, N)** — every product is made using at least one ingredient, up to any number
  - **PRODUCT\_INGREDIENTS (1, 1)** — every product ingredient references exactly one product that it is an ingredient of.
5. **INGREDIENTS & PRODUCT\_INGREDIENTS (1:N)** — Each ingredient can be used in many different products, and each entry in the **PRODUCT\_INGREDIENTS** table refers to a specific ingredient for a specific product.
- **INGREDIENTS (1, N)** — every ingredient is a product ingredient of at least one up to many different products.
  - **PRODUCT\_INGREDIENTS (1, 1)** — every product ingredient references only one ingredient.

## Entity-Relationship (ER) Schemas

### ER Diagram Representation



## Entity-Relationship Schema (using Designer tool in MyPHPAdmin)

